Mitchell Kolb

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Education

Bachelor of Science, Computer Science
Washington State University | Pullman WA

Professional Experience

Software Developer — Pullman, WA
Psychology Department, WSU — Oct 2023 - 2024

- Developed the GSUR Data Automation Tool (Python, Requests, BeautifulSoup) that downloads and analyzes **2,000+** patient health files, cutting a multi-day data process to **under 8 minutes**, reducing manual effort by **95%**.
- Enhanced the EMMA Dashboard by migrating it to updated lab infrastructure and **optimizing legacy code**, boosting platform compatibility, performance, and integration with new health datasets.
- Collaborated with researchers through weekly sprints to refine features and testing, so that the output aligned with the study goals and **resolving 100%** of reported bugs before production.
- Presented the automation tool to **80+** attendees at the 2024 WSU Psychology Symposium, prompting departmental adoption and **enabling a PhD student to defend her dissertation**.

Projects

Living Atlas

- Full-stack web application for environmental data integration, enhancing data accessibility with Python-based endpoints, secure file storage, and scalable real-time collaboration features.
- Developed a high-performance **RESTful** backend using **FastAPI** and **PostgreSQL**, significantly optimizing database queries for faster retrieval and enabling real-time environmental data analysis.
- Implemented **Google Cloud file storage** and **Docker**-based deployments, streamlining data uploads for a seven person team and ensuring secure, scalable access to resources.

Linux EXT2 Filesystem

- Designed and implemented a Linux-compatible EXT2 file system using the C programming language to better understand the operating system's underlying mechanisms.
- Replicated the functionality of the Linux EXT2 architecture by emulating data structures for the SuperBlock, Group Descriptor, Bitmaps, Caches, and Inodes.
- Implemented standard commands like cd, ls, mkdir, rmdir, pwd, link, symlink, etc, along with read/write features (including partial reads and append) to handle more advanced file I/O.

Airline Search Engine

- Engineered a high-performance search engine leveraging **Neo4j**, **PySpark**, and Python to process **67,000+** routes across **7,698** airports, enabling real-time queries and optimal travel routes.
- Implemented a graph-based data model and **MapReduce** operations to streamline route discovery, decreasing query response times by an estimated 50% compared to traditional relational databases.
- Strengthened system scalability by integrating **distributed computing** and **horizontal scaling** strategies, allowing seamless handling of future datasets and increased query loads.

Skills

Programming Languages: Python, SQL, C/C++, HTML
Frameworks: FastAPI, PyQT6, Tkinter, Flask, Pytest, Unittest, GTK
Developer Tools: SwaggerUI, Docker, Jupyter Notebook, Google Cloud Services, VIM, VSCode, Git, PostgreSQL
Libraries: BeautifulSoup, Requests, Selenium, Playwright, PyTorch, Pandas, NumPy, Matplotlib, WRS